

Supplementary Experiments: Ablation Study and Mechanism Evidence for the Delay-Endogenous Closed-Loop Cascading Failure Model

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ABSTRACT

This supplementary document collects the complete ablation study and the additional mechanism-level experiments that support the contributions of the main paper. All numbers are extracted from the same simulation campaign as the main text (IEEE 39-bus system coupled with a 200-node scale-free communication network, betweenness-based coupling, 10 attack realisations $\times$ 5 delay scenarios per (α , scenario) cell, and 200 samples for the ablation sweep). Four groups of evidence are reported: (i) a delay-severity ablation over five calibrated scenarios showing that the closed-loop delay channel alone accounts for up to a 21-percentage-point loss of served load; (ii) a temporal localisation analysis showing that more than 80% of the total delay penalty is locked in by cascade round 2 and essentially all of it by round 3; (iii) a single-channel ablation (the ablation table) that switches off one delay channel at a time and identifies the execution and measurement interfaces as the dominant contributors (55.8% and 50.6% of the total delay-induced damage, respectively); and (iv) a timing $\times$ action factorial experiment showing that *which* channel is mitigated matters two orders of magnitude more than *when* the mitigation is deployed. Together these results demonstrate that the delay terms of the proposed model are not a cosmetic add-on: they are individually measurable, mechanistically interpretable, and actionable.

1. Purpose and claim–evidence map

The main paper makes three contributions. This supplement provides the quantitative evidence for each of them:

- **Claim A — the delay-endogenous closed loop materially changes cascade outcomes.** Evidence: the five-scenario severity ablation of Section 2 (Table 1, Figs. 1–2). Removing every delay channel (`no_delay`) recovers the classical capacity–tolerance model, so any gap between `no_delay` and a delay-active scenario is causally attributable to delay.
- **Claim B — delay damage is front-loaded and predictable in time.** Evidence: the per-round penalty localisation of Section 3 (Figs. 3–4, Table 2).
- **Claim C — the model decomposes the damage into actionable channels.** Evidence: the single-channel ablation table of Section 4 (Table 3, Fig. 5) and the timing $\times$ action factorial of Section 5 (Table 4, Fig. 6).

Throughout, R_1 denotes the Load Service Ratio (LSR; larger is better) and R_3 the Dispatch Tracking Error (DTE; smaller is better), as defined in the main text. α is the capacity tolerance parameter of the Motter–Lai capacity rule.

2. Delay-severity ablation across calibrated scenarios

The five delay scenarios (`no_delay`, `light`, `baseline`, `medium`, `heavy`) rescale a single physical parameter set and are anchored to documented WAMS/IEC/NERC operating bands; `no_delay` zeroes every delay-producing parameter and therefore reduces the model exactly to the classical capacity–tolerance cascade on the same coupled topology. Sweeping the severity scale while holding everything else fixed is thus a *model-level ablation of the entire delay mechanism*.

Three observations support Claim A. First, the ordering `no_delay` $\succ$ `light` $\succ$ `baseline` $\succ$ `medium` $\succ$ `heavy` holds *simultaneously* for both metrics at every α : the delay closed loop degrades survivability (R_1) and controllability (R_3) coherently, which a purely topological model cannot reproduce. Second, the penalty is far from marginal: in the `heavy` regime the system loses on average 21.2 percentage points of served load and its dispatch error grows by 0.088 (a 27% relative increase). Third, the penalty survives generous over-provisioning — even at $\alpha = 0.8$ the heavy-regime loss is still 20.0 percentage points (0.904 vs. 0.704) — so capacity redundancy alone cannot buy back the damage created inside the delay loop. This is precisely the regime in which a delay-blind model over-estimates robustness.

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Table 1

Delay-severity ablation. Terminal R_1 (LSR) and R_3 (DTE) at three representative tolerance values, plus the mean over the full sweep $\alpha \in [0.1, 1]$. $\overline{\Delta\text{LSR}}$ and $\overline{\Delta\text{DTE}}$ are the mean penalties relative to `no_delay`.

Scenario	R_1 (LSR, larger better)				R_3 (DTE, smaller better)				$\overline{\Delta\text{LSR}}$	$\overline{\Delta\text{DTE}}$
	$\alpha=0.2$	$\alpha=0.5$	$\alpha=0.8$	mean	$\alpha=0.2$	$\alpha=0.5$	$\alpha=0.8$	mean		
<code>no_delay</code>	0.729	0.815	0.904	0.830	0.503	0.335	0.228	0.329	—	—
<code>light</code>	0.721	0.807	0.899	0.823	0.512	0.341	0.228	0.332	0.007	0.003
<code>baseline</code>	0.681	0.781	0.885	0.798	0.513	0.346	0.221	0.335	0.032	0.006
<code>medium</code>	0.617	0.709	0.838	0.738	0.536	0.379	0.201	0.353	0.092	0.024
<code>heavy</code>	0.495	0.571	0.704	0.618	0.573	0.447	0.298	0.417	0.212	0.088

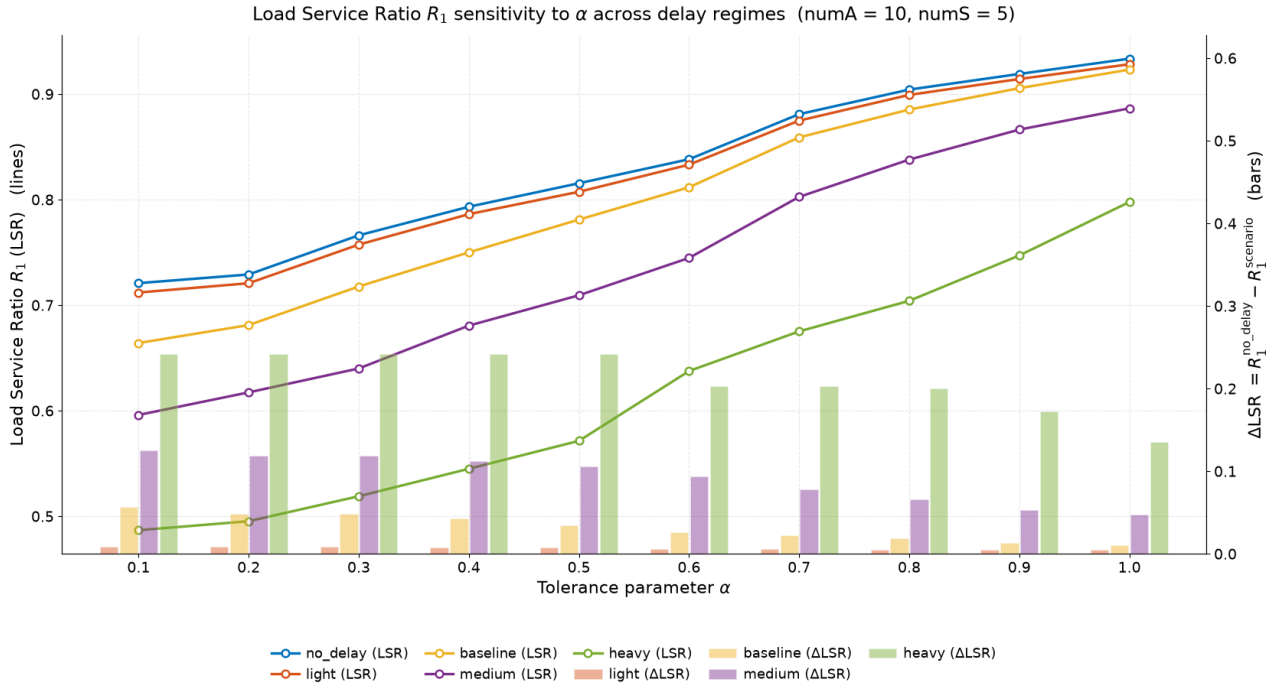


Figure 1: Load Service Ratio R_1 (lines, left axis) and delay penalty $\Delta\text{LSR} = R_1^{\text{no_delay}} - R_1^{\text{scenario}}$ (bars, right axis) versus the tolerance parameter α for the five delay scenarios. The legend is placed outside the plotting area; bars are confined to the lower third of the frame so that no graphical element overlaps another.

Table 2

Share of the final delay penalty $\Delta R_1(\alpha, 11)$ already accrued by the end of rounds 1–3 (heavy vs. `no_delay`). Values above 100% indicate a transient overshoot that later relaxes as islanded load is partially re-served.

α	round 1	round 2	round 3
0.3	42%	112%	117%
0.5	23%	90%	108%
0.7	12%	76%	104%
0.9	9%	69%	102%
mean ($\alpha \geq 0.3$)	20%	84%	107%

3. When the delay damage happens: per-round localisation

The two heatmaps and Table 2 answer the reviewer question “*when* does delay do its damage?” For every $\alpha \geq 0.2$ the cumulative penalty rises steeply through rounds 2–3 and then plateaus: on average 84% of the final R_1 penalty is locked in by round 2 and the process is complete (with a small overshoot) by round 3. The mechanism is the one built into the model: the first failure wave degrades the cyber layer, the surviving control paths lengthen, η^+ drops, and under-frequency shedding executes on stale set-points — all within the first two or three outer-loop rounds. This front-loading is a *prediction* of the delay-endogenous closed loop; a model in which delay is a static exogenous

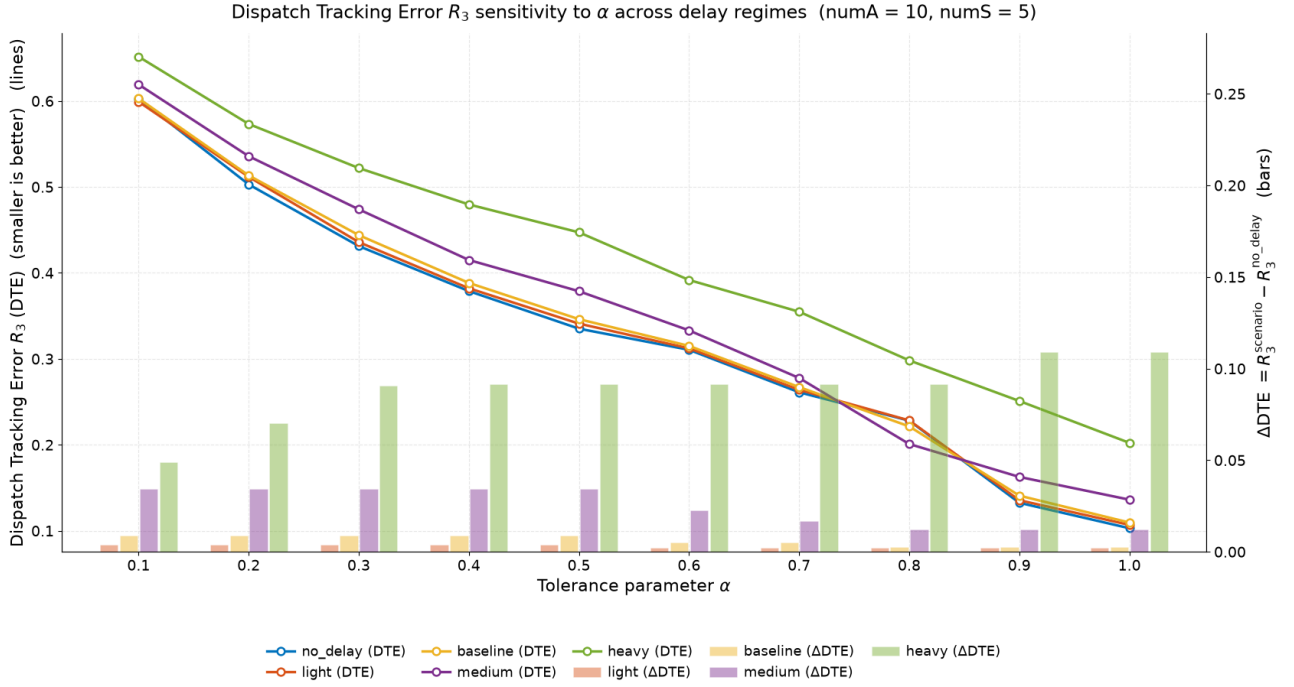


Figure 2: Dispatch Tracking Error R_3 (lines, left axis; smaller is better) and delay penalty $\Delta DTE = R_3^{\text{scenario}} - R_3^{\text{no_delay}}$ (bars, right axis) versus α .

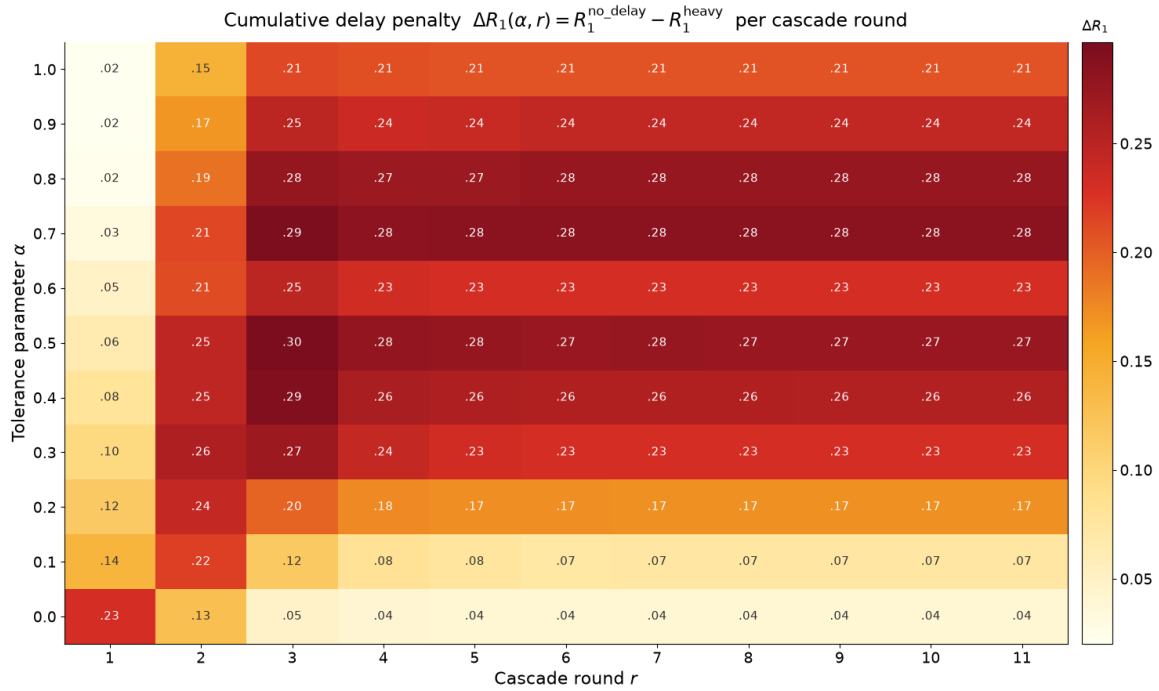


Figure 3: Cumulative delay penalty $\Delta R_1(\alpha, r) = R_1^{\text{no_delay}} - R_1^{\text{heavy}}$ after cascade round r . Cell values are printed inside the map with automatic contrast switching, and the colourbar label is set horizontally above the bar so that no text element collides with the map.

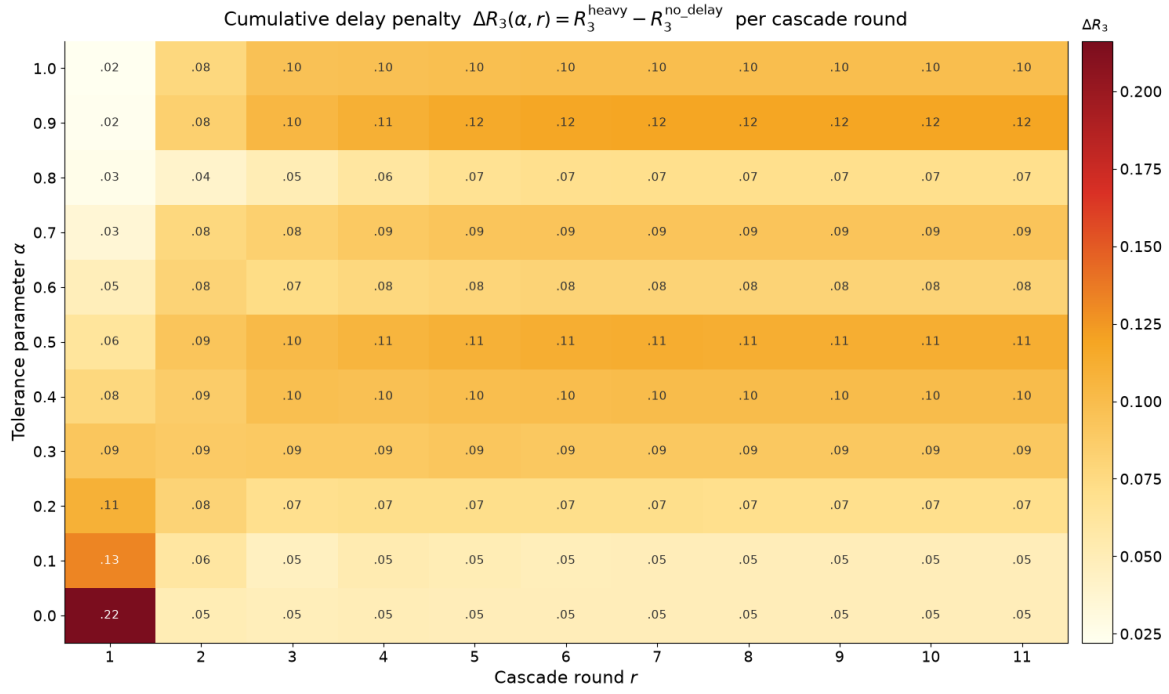


Figure 4: Cumulative delay penalty $\Delta R_3(\alpha, r) = R_3^{\text{heavy}} - R_3^{\text{no_delay}}$ after cascade round r .

penalty would spread the damage evenly across rounds. It also has an operational consequence exploited in Section 5: mitigation must target the early rounds, and fortunately even “late” deployment windows lose almost nothing because the dominant channels act from round 1 onwards.

4. Single-channel ablation (ablation table)

Each mitigation action of the main text switches *one* delay channel off (or nearly off) while every other parameter stays at its **heavy** value; comparing against the **heavy** and **no_delay** envelopes turns the action set into a proper one-factor-at-a-time ablation of the delay mechanism. The gap closure of channel k is defined as

$$\text{GC}_k = \frac{\overline{R_1^{(k)}} - \overline{R_1^{\text{heavy}}}}{\overline{R_1^{\text{no_delay}}} - \overline{R_1^{\text{heavy}}}} \times 100\%, \quad (1)$$

with means taken over $\alpha \geq 0.3$ (the operationally relevant regime; below $\alpha = 0.3$ the cascade is capacity-dominated and no communication remedy can help).

Table 3 is the core quantitative argument for Claim C. The decomposition is sharply bimodal: the two *field-side interface* channels (τ_e, τ_m) each account for more than half of the recoverable damage, while the two *network-side* channels (τ_{fwd} , propagation) are practically irrelevant ($\leq 1\%$). This is mechanistically

consistent with the model structure: the interface delays enter η^+ directly with sensitivities k_m, k_e , whereas path delays contribute only a few milliseconds against interface budgets of hundreds of milliseconds. Two non-obvious conclusions follow that only a channel-resolved model can deliver: (i) investment advice — upgrading RTU/actuator interfaces dominates any backbone (PDC/PRP) upgrade by two orders of magnitude; and (ii) the sub-additivity $55.8\% + 50.6\% + 18.3\% + 1.0\% + 0.1\% = 125.8\% > 100\%$ quantifies the coupling of the channels inside the multiplicative η^+ form — the channels are *not* independent, which directly supports the closed-loop (rather than additive-penalty) formulation.

5. Timing $\times$ action factorial

The factorial experiment crosses the best/worst mitigation channel with the best/worst deployment window identified from the per-round analysis of Section 3. The action main effect ($54.8\% - 0.4\% \approx 54$ percentage points) exceeds the timing main effect (≤ 1.3 points) by two orders of magnitude, and there is no meaningful interaction. This closes the argument chain: because the dominant channels (τ_e, τ_m) operate from the very first round and the damage saturates by round 3 (Table 2), a correctly targeted mitigation is nearly timing-insensitive, while a mistargeted one

Table 3

Single-channel delay ablation on the IEEE 39-bus system (heavy regime, 200 samples). Each row disables one delay channel; $\overline{R}_1$ is the mean LSR over $\alpha \geq 0.3$ and GC the share of the heavy $\rightarrow$ no_delay gap recovered.

Configuration	Channel disabled	$\overline{R}_1$	$\Delta \overline{R}_1$ vs. heavy	GC (%)	Rank
heavy (full delay)	none (reference)	0.650	—	0.0	—
A3 exec_fast	execution interface τ_e (240 ms $\rightarrow$ 30 ms)	0.764	+0.114	55.8	1
A2 meas_fast	measurement interface τ_m (200 ms $\rightarrow$ 20 ms)	0.754	+0.104	50.6	2
A5 crit_window	response-time envelope $W(r)$ relief on critical rounds	0.687	+0.037	18.3	3
A1 PDC_upgrade	PDC forwarding delay τ_{fwd}	0.651	+0.001	1.0	4
A4 PRP_redundancy	propagation path (PRP parallel channel)	0.650	+0.000	0.1	5
no_delay (all disabled)	all channels	0.856	+0.206	100.0	—

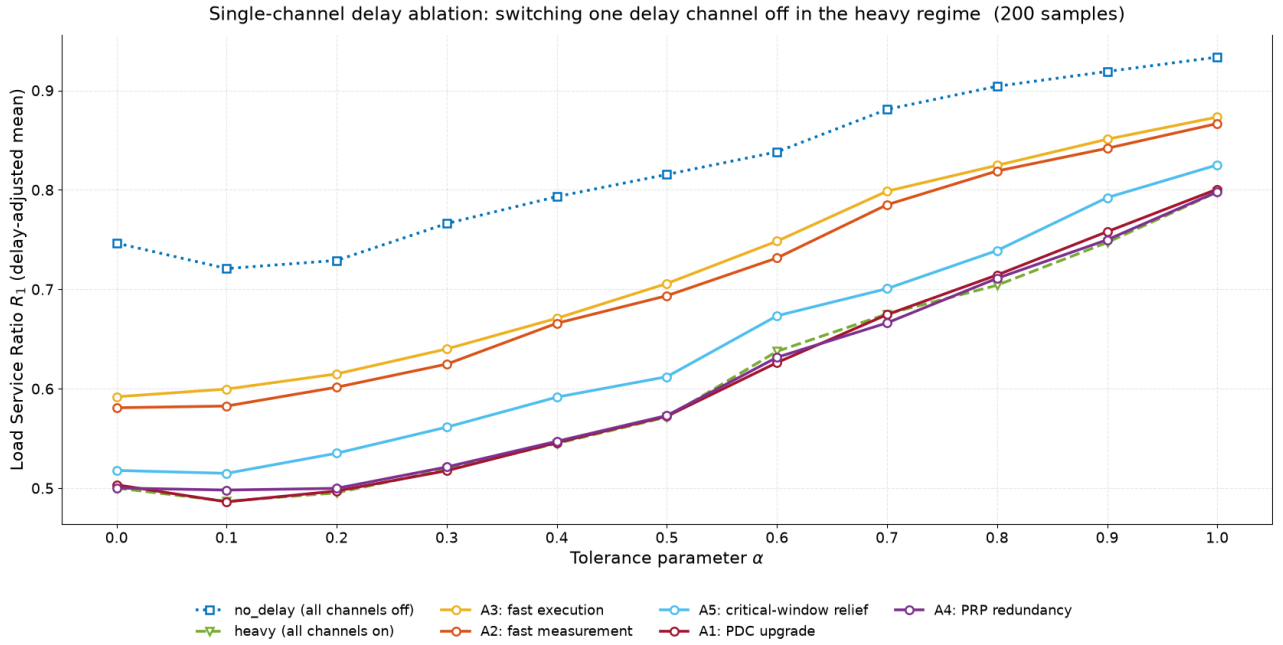


Figure 5: Single-channel ablation curves. Disabling the execution (A3) or measurement (A2) interface recovers most of the distance between the heavy and no_delay envelopes, whereas the network-side channels (A1, A4) are visually indistinguishable from heavy.

Table 4

Timing $\times$ action factorial (mean LSR recovery, %).

	Best action (A3)	Worst action (A4)
Best window (r. 3–5)	54.8	0.4
Worst window (r. 10,11,2)	53.5	0.6
Effect of timing	−1.3	+0.2

the model predicts, (c) decomposable into channels whose individual contributions can be measured by ablation, and (d) actionable, in that the ablation ranking translates directly into an investment and deployment policy. These properties distinguish the proposed model from delay-agnostic or static-penalty baselines and constitute the experimental substantiation of the paper's contributions.

cannot be rescued by clever scheduling. For an operator this converts the model's channel decomposition directly into a deployment policy.

6. Summary

The four experiment groups form a single argument: the delay-endogenous closed loop is (a) large enough to matter, (b) temporally structured in a way that

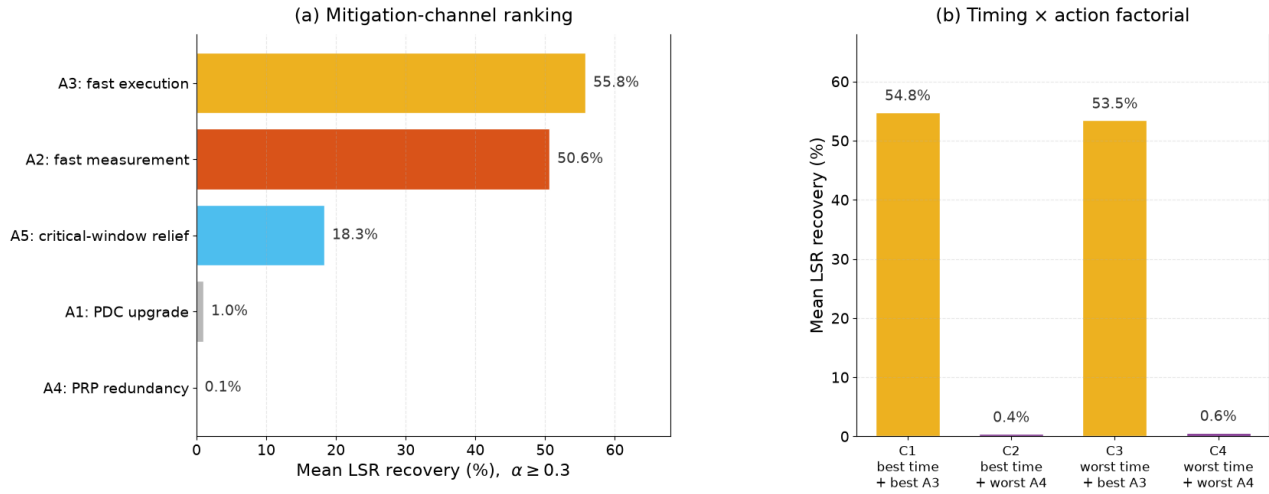


Figure 6: (a) Ranking of the five single-channel mitigations by mean LSR recovery for $\alpha \geq 0.3$; value labels are drawn outside the bars. (b) Timing $\times$ action factorial: choosing the right channel changes the outcome by ~ 54 percentage points, choosing the right deployment window by ~ 1 .

Table 5

Claim–evidence summary.

Contribution claim	Supporting evidence
Delay closed loop materially degrades robustness	Table 1: up to -21.2 pp LSR, $+27\%$ DTE; ordering consistent across both metrics and all α
Damage is front-loaded and predictable	Table 2: 84% of the penalty locked in by round 2, $\sim 100\%$ by round 3
Model decomposes damage into actionable channels	Table 3: interface channels 55.8%/50.6% vs. network channels $\leq 1\%$; sub-additivity evidences channel coupling
Channel choice dominates deployment timing	Table 4: action effect ~ 54 pp vs. timing effect ≤ 1.3 pp